

The main objective of this phase is to clearly identify and understand the business problem addressed by the system. The project focuses on solving challenges faced in modern agriculture such as poor crop selection, inefficient resource utilization, changing environmental conditions, and reduced farming productivity. Farmers often struggle to determine the most suitable crop based on soil nutrients and climate conditions, which can lead to financial losses and low agricultural yield.

This phase involves analyzing the agricultural domain and identifying the important environmental and soil parameters that directly affect crop growth and productivity. Key factors such as Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH value, rainfall, and seasonal conditions are studied to understand their impact on crop recommendation and agricultural optimization.

The system aims to provide an intelligent and data-driven solution that assists farmers, agricultural researchers, agribusiness organizations, and policymakers in making better farming decisions. By integrating machine learning techniques and predictive analytics, the platform can recommend the most suitable crops for specific environmental conditions and help improve agricultural efficiency, sustainability, and productivity.

This phase also includes understanding the practical applications, business requirements, user needs, and expected outcomes of the system. The knowledge gathered during this stage forms the foundation for data collection, machine learning model development, application design, and deployment of the agricultural recommendation system.